

Assignment 7

fredag 28. februar 2025

16:41

Problem 1:

a) Three phase systems deliver more power and the power is constant

b) Phase voltage: voltage between phase and neutral
Line to line: voltage between two lines

$$c) V_{bn}(t) = \sqrt{2} V_{rms} \cos(\omega t - 120^\circ)$$

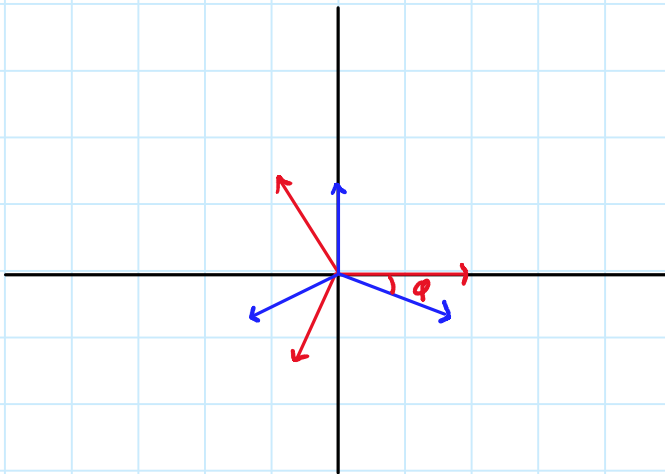
$$V_{cn}(t) = \sqrt{2} V_{rms} \cos(\omega t - 240^\circ)$$

$$i_b(t) = \sqrt{2} I_{rms} \cos(\omega t + \varphi - 120^\circ)$$

$$i_c(t) = \sqrt{2} I_{rms} \cos(\omega t + \varphi - 240^\circ)$$

$$d) \begin{aligned} v_{an}(t) &: \sqrt{2} V_{rms} e^{j \cdot 0} \\ v_{bn}(t) &: \sqrt{2} V_{rms} e^{j \frac{2\pi}{3}} \\ v_{cn}(t) &: \sqrt{2} V_{rms} e^{j \frac{4\pi}{3}} \end{aligned}$$

$$\begin{aligned} i_a(t) &: \sqrt{2} I_{rms} e^{j \cdot \varphi} \\ i_b(t) &: \sqrt{2} I_{rms} e^{j(\varphi - \frac{2\pi}{3})} \\ i_c(t) &: \sqrt{2} I_{rms} e^{j(\varphi - \frac{4\pi}{3})} \end{aligned}$$



$$e) \begin{aligned} V_{ab} &= V_{an} - V_{bn} \\ V_{bc} &= V_{bn} - V_{cn} \end{aligned}$$

$$e) V_{ab} = V_{an} - V_{bn}$$

$$V_{bc} = V_{bn} - V_{cn}$$

$$V_{ab} = V_{an} - V_{bn} = V_{rms} (e^{j0^\circ} - e^{-j120^\circ})$$

$$\rightsquigarrow = \sqrt{3} V_{rms} e^{j30^\circ}$$

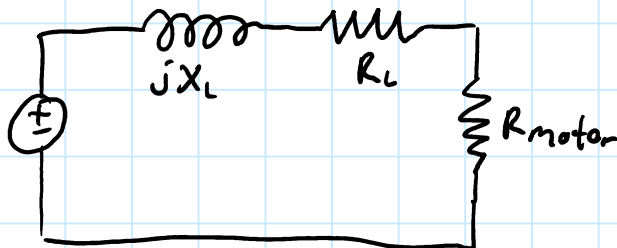
f) The 3-phase system is in a Wye (star) connection with a neutral point.

Your phone is connected between one phase and neutral.

$$\text{Phase to neutral voltage: } V = \frac{V_{LL}}{\sqrt{3}} = \frac{400V}{\sqrt{3}} = 230,9V$$

Problem 2:

$$a) V_{an} = \frac{V_{ab}}{\sqrt{3}} = 230,9V$$



$$b) i = \frac{V_{an}}{jX_L + R_{motor} + R_L}$$

$$P_{motor} = \frac{V_{ab}^2 \cdot 0,5}{|0,025j + 0,5|^2} = 306851W$$

$$= 409hp$$

$$c) W = \frac{1}{2} m v^2 = 771600J$$

$$P = 306851W$$

$$P = \frac{W}{t} \Rightarrow t = \frac{W}{P}$$

$$t = \frac{771600J}{306851W} \approx 2,5s$$

$$t = \frac{1716000}{306851W} \approx 2,5s$$

$$d) \frac{70 kWh}{0,8 \cdot 306851W} = 1026s$$

Problem 3:

$$a) \frac{N_1}{N_2} = \frac{V_s}{V_i}, \quad V_o = 0,91 V_i$$

Rectifier output: $V_o = 5V$

$$V_i = \frac{V_o}{0,9} = 5,56V$$

$$\frac{N_1}{N_2} = \frac{240V}{5,56V} \approx 43,2$$

$N_1 : N_2$ is approx 43:1

$$b) V_i = \frac{4,5}{0,9} = 5$$

$$V_1 = \frac{5}{\frac{1}{41}} = 216 \Rightarrow -i_s R + V_s = 216$$

$$i_s = 0,024A$$

$$i_o = \frac{0,024A}{\frac{0,0231}{0,9}} = 1,15A$$

$$c) \text{ Loss: } R i_s^2 = 0,57W$$

$$\frac{0,57W}{240V \cdot 24 \cdot 10^{-6}A} = 0,1$$

Problem 4:

a) Stator: Stationary windings

Rotor: Something that rotates

b) Field winding: coils that create the primary magnetic field in the machine.

Armature: Where the main voltage is induced.

c) Permanent magnets take the place of the field winding.

d) AC-machines:

- Synchronous

- Induction

Synchronous machines are used in power plants. They have const. speed.

Part 2

a) $Z_{rns} = Z_L + R_2 + j\omega L_2 = (160 + 120j) \Omega$

$$Z_{Lhs} = Z_s + R_1 + j\omega L_1 = (18,34 + 48,75j) \Omega$$

b) Impedance reflected into primary:

$$\frac{M^2 \omega^2}{Z} = 11,66 - 8,75j$$

c) $Z_{ab} = Z_{total} - Z_s$

$$Z = \frac{V_s}{I} - Z_s = \frac{75}{10} - (18,34 + 48,75j) = 7,5 - 48,75j$$

$$Z_{ab} = \frac{V_s}{I_1} - Z_s = \frac{75}{0,9 - 1,2j} - (10 + 12,75j)$$
$$= 20 + 27,25j$$